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RT-DETRv2: Improved Baseline with Bag-of-Freebies for Real-Time Detection Transformer

Technical Report

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Abstract

In this report, we present RT-DETRv2, an improved Real-Time DEtection TRansformer (RT-DETR). RT-DETRv2 builds upon the previous state-of-the-art real-time detector, RT-DETR, and opens up a set of bag-of-freebies for flexibility and practicality, as well as optimizing the training strategy to achieve enhanced performance. To improve the flexibility, we suggest setting a distinct number of sampling points for features at different scales in the deformable attention to achieve selective multi-scale feature extraction by the decoder. To enhance practicality, we propose an optional discrete sampling operator to replace the `grid_sample` operator that is specific to RT-DETR compared to YOLOs. This removes the deployment constraints typically associated with DETRs. For the training strategy, we propose dynamic data augmentation and scale-adaptive hyperparameters customization to improve performance without loss of speed. Source code and pre-trained models will be available at https://github.com/lyuwenyu/RT-DETR.

1 Introduction

Object detection is a fundamental vision task that involves identifying and localizing objects in an image. Among them, real-time object detection is an important field and has a wide range of applications, such as autonomous driving (Atakishiyev et al. [2024]). With the development of the last few years, YOLO detectors (Redmon and Farhadi [2017, 2018], Bochkovskiy et al. [2020], Glenn. [2022], Xu et al. [2022], Li et al. [2023], Wang et al. [2023], Glenn. [2023], Wang et al. [2024a,b]) are without doubt the most prestigious framework in this field. The reason for this is the reasonable balance achieved by the YOLO detectors.

The advent of RT-DETR (Zhao et al. [2024]) opens up a new technological avenue for real-time object detection, breaking the dependency on the YOLO in this field. RT-DETR proposes an efficient hybrid encoder to replace the vanilla Transformer encoder in DETR (Carion et al. [2020]), which significantly improves the inference speed by decoupling the intra-scale interaction and cross-scale fusion of multi-scale features. To further improve the performance, RT-DETR proposes the uncertainty-minimal query selection, which provides high-quality initial queries to the decoder by explicitly optimizing the uncertainty. Moreover, RT-DETR provides a wide range of detector sizes and supports flexible speed tuning to accommodate various real-time scenarios without retraining. RT-DETR represents a novel, end-to-end, real-time detector that marks a significant advancement for the DETR family.

In this report, we present RT-DETRv2, an improved real-time detection Transformer. This work is built upon the recent RT-DETR and opens up a set of bag-of-freebies for flexibility and practicality within the DETR family, as well as optimizing the training strategy to achieve enhanced performance. Specifically, RT-DETRv2 suggests setting a distinct number of sampling points for features at different scales within the deformable attention module to achieve selective multi-scale feature extraction by the decoder. In the realm of enhancing practicality, RT-DETRv2 provides an optional discrete sampling operator to replace the original `grid_sample` operator, which is specific to DETRs, thus eliminating the deployment constraints typically associated with detection Transformers. Furthermore, RT-DETRv2 optimizes the training strategy, including dynamic data augmentation and scale-adaptive hyperparameters customization, with the objective of improving performance without loss of speed. The results demonstrate that RT-DETRv2 provides an improved baseline with bag-of-freebies for RT-DETR, increases the flexibility and practicality, and the proposed training strategies optimize the performance and training cost.

RT-DETRv2：基于即用型优化策略改进的实时检测Transformer基线模型

技术报告

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摘要

在本报告中，我们提出了RT-DETRv2，一种改进的实时检测变换器（RT-DETR）。RT-DETRv2基于先前最先进的实时检测器RT-DETR构建，并引入了一系列即插即用的优化方案以提升灵活性和实用性，同时优化了训练策略以实现更强的性能。为提高灵活性，我们建议在可变形注意力机制中为不同尺度的特征设置不同数量的采样点，从而使解码器器能够实现选择性的多尺度特征提取。为增强实用性，我们提出了一种可选的离散采样算子，以替代RT-DETR相较于YOLO系列特有的grid_sample算子。此举消除了DETR系列模型通常面临的部署限制。在训练策略方面，我们提出了动态数据增强和尺度自适应超参数定制方法，在不损失速度的前提下提升性能。源代码与预训练模型将在https://github.com/lyuwenyu/RT-DETR发布。

1 引言

目标检测是一项基础的视觉任务，涉及识别并定位图像中的物体。其中，实时目标检测是一个重要领域，具有广泛的应用，例如自动驾驶（Atakishiyev等人[2024]）。随着近几年的发展，YOLO检测器（Redmon和Farhadi[2017, 2018]、Bochkovskiy等人[2020]、Glenn[2022]、Xu等人[2022]、Li等人[2023]、Wang等人[2023]、Glenn[2023]、Wang等人[2024a,b]）无疑是该领域最具声望的框架。这得益于YOLO检测器所实现的合理平衡。

RT-DETR（Zhao等人[2024]）的出现为实时目标检测开辟了新的技术路径，打破了该领域对YOLO的依赖。RT-DETR提出了一种高效的混合编码器，以替代DETR（Carion等人[2020]）中的原始Transformer编码器，通过解耦多尺度特征的尺度内交互与跨尺度融合，显著提升了推理速度。为进一步提升性能，RT-DETR提出了不确定性最小化查询选择机制，通过显式优化不确定性为解码器提供高质量的初始查询。此外，RT-DETR提供了多种检测器尺寸，并支持灵活的速度调节，无需重新训练即可适应多样化的实时场景。RT-DETR代表了一种新颖的端到端实时检测器，标志着DETR系列的重要进展。

在本报告中，我们提出了RT-DETRv2，一种改进的实时检测Transformer。这项工作基于最新的RT-DETR，为DETR系列开辟了一套提升灵活性与实用性的免费技巧包，并通过优化训练策略以实现更强的性能。具体而言，RT-DETRv2建议在可变形注意力模块中为不同尺度的特征设置不同数量的采样点，从而使解码器能够实现选择性的多尺度特征提取。在提升实用性方面，RT-DETRv2提供了一个可选的离散采样算子，以替代原本专用于DETR的grid_sample算子，从而消除了检测Transformer通常面临的部署限制。此外，RT-DETRv2优化了训练策略，包括动态数据增强和尺度自适应超参数定制，旨在不损失速度的前提下提升性能。实验结果表明，RT-DETRv2为RT-DETR提供了一个带有免费技巧包的改进基线，增强了灵活性与实用性，且所提出的训练策略优化了性能与训练成本。

2 Method

The framework of RT-DETRv2 remains the same as RT-DETR, with only modifications to the deformable attention module of the decoder.

2.1 Framework

Distinct number of sampling points for different scales. Current DETRs utilize the deformable attention module (Zhu et al. [2020]) to alleviate the high computational overhead caused by the long sequence of inputs composed of multi-scale features. The RT-DETR decoder retains this module, which defines the same number of sampling points at each scale. We argue that this constraint ignores the intrinsic differences in features at different scales and limits the feature extraction capability of the deformable attention module. Therefore, we propose to set distinct numbers of sampling points for different scales to achieve more flexible and efficient feature extraction.

Discrete sampling. To improve the practicality of the RT-DETR and to make it available everywhere. We focus on comparing the deployment requirements of YOLOs and RT-DETR, where the RT-DETR-specific `grid_sample` operator limits its broad applicability. Therefore, we propose an optional `discrete_sample` operator to replace the `grid_sample`, thus removing the deployment constraints of RT-DETR. Specifically, we perform a rounding operation on the predicted sampling offsets, omitting the time-consuming bilinear interpolation. However, the rounding operation is non-differentiable, so we turn off the gradient of the parameters used to predict the sampling offsets. In practice, we first employ the `grid_sample` operator for training and then replace it with the `discrete_sample` operator for fine-tuning. For inference and deployment, the model employs the `discrete_sample` operator.

2.2 Training Scheme

Dynamic data augmentation. To equip the model with robust detection performance, we propose the dynamic data augmentation strategy. Considering the poor generalizability of the detector in the early training period, we apply stronger data augmentation, while in the later training period we decrease its level to adapt the detector to the detection of the target domain. Specifically, we maintain the RT-DETR data augmentation in the early period, while turning off `RandomPhotometricDistort`, `RandomZoomOut`, `RandomIoUCrop`, and `MultiScaleInput` in the last two epochs.

Scale-adaptive hyperparameters customization. We also observe that the scaled RT-DETRs of different sizes are trained with the same optimizer hyperparameters, resulting in their sub-optimal performance. Therefore, we propose scale-adaptive hyperparameters customization for scaled RT-DETRs. Considering that the pre-trained backbone for light detector (*e.g.*, ResNet18 (He et al. [2016])) has lower feature quality, we increase its learning rate. On the contrary, the pre-trained backbone with large detector (*e.g.*, ResNet101 (He et al. [2016])) has higher feature quality and we decrease its learning rate.

3 Experiment

3.1 Implementation Details

As with RT-DETR, we use ResNet (He et al. [2016]) pretrained on ImageNet as the backbone and train RT-DETRv2 with the AdamW (Loshchilov and Hutter [2018]) optimizer with a batch size of 16 and apply the exponential moving average (EMA) with $ema\_decay = 0.9999$. For the optional discrete sampling, we first pre-train $6\times$ with the `grid_sample` operator and then fine-tune $1\times$ with the `discrete_sample` operator. For scale-adaptive hyperparameters customization, the hyperparameters are shown in Tab. 1, where *lr* represents the learning rate.

Table 1: The hyperparameters of RT-DETRv2.

Model	**Backbone**	$lr_{backbone}$	lr_{det}
RT-DETRv2-S	ResNet18	1e-4	1e-4
RT-DETRv2-M	ResNet34	5e-5	1e-4
RT-DETRv2-L	ResNet50	1e-5	1e-4
RT-DETRv2-X	ResNet101	1e-6	1e-4

3.2 Evaluation

RT-DETRv2 is trained on COCO (Lin et al. [2014]) `train2017` and validated on COCO `val2017` dataset. We report the standard AP metrics (averaged over uniformly sampled IoU thresholds ranging from $0.50-0.95$ with a step size of 0.05), and AP^{val}_{50} commonly used in real scenarios.

2 方法

RT-DETRv2的框架与RT-DETR保持一致，仅对解码器的可变形注意力模块进行了修改。

2.1 框架

不同尺度的采样点数量不同。当前的DETR模型采用可变形注意力模块（Zhu等人[2020]）来缓解由多尺度特征构成的长输入序列带来的高计算开销。RT-DETR解码器保留了该模块，其在每个尺度上定义了相同数量的采样点。我们认为这种约束忽略了不同尺度特征的内在差异，限制了可变形注意力模块的特征提取能力。因此，我们建议为不同尺度设置不同数量的采样点，以实现更灵活高效的特征提取。

离散采样。为了提高RT-DETR的实用性并使其能够随处可用，我们重点比较了YOLO系列与RT-DETR的部署需求，其中RT-DETR特有的grid_sample算子限制了其广泛适用性。为此，我们提出了一种可选的discrete_sample算子来替代grid_sample，从而消除RT-DETR的部署限制。具体而言，我们对预测的采样偏移量执行取整操作，省去了耗时的双线性插值。然而取整操作不可微分，因此我们关闭了用于预测采样偏移量的参数梯度。在实际应用中，我们首先使用grid_sample算子进行训练，随后在微调阶段替换为discrete_sample算子。在推理与部署时，模型均采用discrete_sample算子。

2.2 训练方案

动态数据增强。为了使模型具备鲁棒的检测性能，我们提出了动态数据增强策略。考虑到检测器在训练初期的泛化能力较差，我们采用更强的数据增强方法；而在训练后期，我们降低增强强度，使检测器适应目标域的检测任务。具体而言，我们在训练前期保持RT-DETR的数据增强设置，而在最后两个训练周期中关闭RandomPhotometricDistort、RandomZoomOut、RandomIoUCrop和MultiScaleInput增强模块。

尺度自适应超参数定制。我们还观察到，不同尺寸的缩放RT-DETR模型使用相同的优化器超参数进行训练，导致其性能未达最优。因此，我们针对缩放RT-DETR模型提出了尺度自适应超参数定制方案。考虑到轻量检测器（*e.g.*，ResNet18（He等人[2016]））的预训练主干网络特征质量较低，我们提高了其学习率。相反，大型检测器（*e.g.*，ResNet101（He等人[2016]））的预训练主干网络具有更高的特征质量，因此我们降低了其学习率。

3 实验

3.1 实现细节

与RT-DETR相同，我们使用在ImageNet上预训练的ResNet（He等人[2016]）作为骨干网络，并采用AdamW优化器（Loshchilov和Hutter[2018]）训练RT-DETRv2，批大小为16，同时应用指数移动平均（EMA），其参数为ema_decay =0.9999。对于可选的离散采样，我们首先使用grid_sample算子预训练6×，然后使用discrete_sample算子微调1×。对于尺度自适应超参数定制，相关超参数如表1所示，其中lr代表学习率。

表1：RT-DETRv2的超参数。

Model	**Backbone**	$lr_{backbone}$	lr_{det}
RT-DETRv2-S	ResNet18	1e-4	1e-4
RT-DETRv2-M	ResNet34	5e-5	1e-4
RT-DETRv2-L	ResNet50	1e-5	1e-4
RT-DETRv2-X	ResNet101	1e-6	1e-4

3.2 评估

RT-DETRv2在COCO（Lin等人[2014]）train2017数据集上训练，并在COCO val2017数据集上进行验证。我们报告了标准AP指标（在0.50至0.95范围内以0.05为步长均匀采样的IoU阈值上的平均值），以及实际场景中常用的AP_{50}^{val}。

3.3 Results

The comparison with RT-DETR(Zhao et al. [2024]) is shown in Tab. 2. RT-DETRv2 outperforms RT-DETR at different scales of detectors without loss of speed.

Table 2: **Comparison of RT-DETR and RT-DETRv2.** The FPS is reported on T4 GPU with TensorRT FP16. For evaluation, all input sizes are fixed on 640×640.

Model	Backbone	Dataset	#Params (M)	$\textbf{FPS}_{bs=1}$	$\textbf{AP}^{val}$	$\textbf{AP}^{val}_{50}$
RT-DETR-S	ResNet18	COCO	20	217	46.5	63.8
RT-DETR-M	ResNet34	COCO	31	161	48.9	66.8
RT-DETR-M*	ResNet50	COCO	36	145	51.3	69.6
RT-DETR-L	ResNet50	COCO	42	108	53.1	71.3
RT-DETR-X	ResNet101	COCO	76	74	54.3	72.7
RT-DETRv2-S	ResNet18	COCO	20	217	47.9 (↑ **1.4**)	64.9 (↑ **1.1**)
RT-DETRv2-M	ResNet34	COCO	31	161	49.9 (↑ **1.0**)	67.5 (↑ **0.7**)
RT-DETRv2-M*	ResNet50	COCO	36	145	51.9 (↑ **0.6**)	69.9 (↑ **0.3**)
RT-DETRv2-L	ResNet50	COCO	42	108	53.4 (↑ **0.3**)	71.6 (↑ **0.3**)
RT-DETRv2-X	ResNet101	COCO	76	74	54.3 (↑ **0.0**)	72.8 (↑ **0.1**)

3.4 Ablations

Ablation on sampling points. We perform an ablation study on the total number of sampling points of the `grid_sample` operator. The total number of sampling points is calculated as `num_head`×`num_point`×`num_query`×`num_decoder`, where `num_point` represents the sum of sampling points for each scale feature in each grid. The results show that reducing the number of sampling points does not cause a significant degradation in the performance, *cf.* Tab. 3. This means that practical application is unlikely to be affected in most industrial scenarios.

Table 3: Ablation on sampling points.

Model	Sampling method	#Points	$\textbf{AP}^{val}$	$\textbf{AP}^{val}_{50}$
RT-DETRv2-S	`grid_sample`	86,400	47.9	64.9
RT-DETRv2-S	`grid_sample`	64,800	47.8	64.8 (↓ **0.1**)
RT-DETRv2-S	`grid_sample`	43,200	47.7	64.7 (↓ **0.2**)
RT-DETRv2-S	`grid_sample`	21,600	47.3	64.3 (↓ **0.6**)

Ablation on discrete sampling. We then remove the `grid_sample` and replace it with `discrete_sample` for the ablation. The results show that this operation does not cause a noticeable reduction in AP^{val}_{50}, but does eliminate the deployment constraints of the DETRs, *cf.* Tab. 4.

Table 4: Ablation on discrete sampling.

Model	Backbone	Sampling method	$\textbf{AP}^{val}$	$\textbf{AP}^{val}_{50}$
RT-DETRv2-S	ResNet18	`discrete_sample`	47.4	64.8 (↓ **0.1**)
RT-DETRv2-M	ResNet34	`discrete_sample`	49.2	67.1 (↓ **0.4**)
RT-DETRv2-M*	ResNet50	`discrete_sample`	51.4	69.7 (↓ **0.2**)
RT-DETRv2-L	ResNet50	`discrete_sample`	52.9	71.3 (↓ **0.3**)

4 Conclusion

In this report, we propose RT-DETRv2, an improved real-time detection Transformer. RT-DETRv2 opens up a set of bag-of-freebies to increase the flexibility and practicality of RT-DETR, optimizing the training strategy to achieve enhanced performance without loss of speed. We hope that this report will provide insights for the DETR family and broaden the scope of RT-DETR applications.

3.3 结果

与RT-DETR（Zhao等人[2024]）的对比结果如表2所示。RT-DETRv2在不同检测器规模下均优于RT-DETR，且未损失速度。

表2：RT-DETR与RT-DETRv2的对比。FPS数据基于T4 GPU并使用TensorRT FP16测得。评估时，所有输入尺寸固定为640 $\times$ 640。

Model	**Backbone**	**Dataset**	**#Params (M)**	**FPS$_{bs=1}$**	**AP^{val}**	**AP$^{val}_{50}$**
RT-DETR-S	ResNet18	COCO	20	217	46.5	63.8
RT-DETR-M	ResNet34	COCO	31	161	48.9	66.8
RT-DETR-M*	ResNet50	COCO	36	145	51.3	69.6
RT-DETR-L	ResNet50	COCO	42	108	53.1	71.3
RT-DETR-X	ResNet101	COCO	76	74	54.3	72.7
RT-DETRv2-S	ResNet18	COCO	20	217	47.9 ($\uparrow$ **1.4**)	64.9 ($\uparrow$ **1.1**)
RT-DETRv2-M	ResNet34	COCO	31	161	49.9 ($\uparrow$ **1.0**)	67.5 ($\uparrow$ **0.7**)
RT-DETRv2-M*	ResNet50	COCO	36	145	51.9 ($\uparrow$ **0.6**)	69.9 ($\uparrow$ **0.3**)
RT-DETRv2-L	ResNet50	COCO	42	108	53.4 ($\uparrow$ **0.3**)	71.6 ($\uparrow$ **0.3**)
RT-DETRv2-X	ResNet101	COCO	76	74	54.3 ($\uparrow$ **0.0**)	72.8 ($\uparrow$ **0.1**)

3.4 消融实验

采样点消融实验。我们对grid_sample操作的总采样点数进行了消融研究。总采样点数计算公式为num_head$\times$ num_point$\times$num_query$\times$ num_decoder，其中num_point代表每个网格中各尺度特征采样点数量之和。结果表明，减少采样点数量不会导致性能显著下降，*cf.* 表3。这意味着在大多数工业场景中，实际应用不太可能受到影响。

表3：采样点消融实验。

Model	**Sampling method**	**#Points**	**AP^{val}**	**AP$^{val}_{50}$**
RT-DETRv2-S	`grid_sample`	86,400	47.9	64.9
RT-DETRv2-S	`grid_sample`	64,800	47.8	64.8 ($\downarrow$ **0.1**)
RT-DETRv2-S	`grid_sample`	43,200	47.7	64.7 ($\downarrow$ **0.2**)
RT-DETRv2-S	`grid_sample`	21,600	47.3	64.3 ($\downarrow$ **0.6**)

关于离散采样的消融实验。我们随后移除了grid_sample并用discrete_sample进行替换以进行消融。结果显示，这一操作并未导致AP$^{val}_{50}$显著下降，但确实消除了DETRs的部署限制，*cf.*见表4。

表4：离散采样消融实验。

Model	**Backbone**	**Sampling method**	**AP^{val}**	**AP$^{val}_{50}$**
RT-DETRv2-S	ResNet18	`discrete_sample`	47.4	64.8 ($\downarrow$ **0.1**)
RT-DETRv2-M	ResNet34	`discrete_sample`	49.2	67.1 ($\downarrow$ **0.4**)
RT-DETRv2-M*	ResNet50	`discrete_sample`	51.4	69.7 ($\downarrow$ **0.2**)
RT-DETRv2-L	ResNet50	`discrete_sample`	52.9	71.3 ($\downarrow$ **0.3**)

4 结论

在本报告中，我们提出了RT-DETRv2，一种改进的实时检测Transformer。RT-DETRv2开放了一系列“免费增益”方案，以提升RT-DETR的灵活性与实用性，通过优化训练策略在保持速度的同时实现更强的性能。我们希望本报告能为DETR系列模型提供新思路，并拓展RT-DETR的应用范围。

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